Game States

- In each game, there are 18 (non-terminal) states.
  - 0-0, 0-15, 0-30, 0-40
  - 15-0, 15-15, 15-30, 15-40
  - 30-0, 30-15, 30-30, 30-40
  - 40-0, 40-15, 40-30, 40-40 (deuce)
  - Advantage server (ad-in), Advantage receiver (ad-out)

Step 1: Game Winning Probability on Each Game State

- Calculate the probability of winning the current game (not match) for each game state s.
  - With point-by-point datasets in the Match Charting Project (MCP): https://github.com/JeffSackmann/tennis_MatchChartingProject
  - $V(s) = P(\text{server wins game } | s) = \dfrac{\text{\#of times server eventually won game from state } s}{\text{\# of times state } s \text{ occured}}$
    - Focus on the server's perspective here.
    - This is a game-level metric.

Step 2: Average Value of each Event Type across all Game States

- We consider the following event types:
  - Service ace or service winner
    - These are not separated in the meantime.
    - Service ace: untouched by the returner
    - Service winner: touched by the returner, but the return is not in play
  - Double fault
  - Return error (only on return shot)
  - Winner (rally winner incl. return winner)
  - Forced error drawn
  - Unforced error

- This list of event types will be extended.

- Given a set of event types: $\mathcal{E} = \{e_1, e_2, \dots e_i, \dots e_N\}$
- For every point $j = 1, 2, \dots, M$ in the dataset,
  - Identify the pre-point game state $s_{j,\text{before}}$
  - Identify the post-point game state $s_{j,\text{after}}$
  - Identify the point-ending event type $E(j) \in \mathcal{E}$
  - Compute $\Delta V_j = V(s_{j,\text{after}}) - V(s_{j,\text{before}})$ if the server recorded $E(j)$
  - Compute $\Delta V_j = V(s_{j,\text{before}}) - V(s_{j,\text{after}})$ if the returner recorded $E(j)$
    - ΔV_j indicates how much $E(j)$ changed game-winning probability (i.e. how valuable/important $E(j)$ is).
  - Assign ΔV_j to the event type $E(j)$
- Then, compute the averageΔV_j for each event type
  - Formally, $w(e_i) = \frac{1}{N_i} \sum_{j:E(j)=e_i} \Delta V_j$
    - $w(e_i)$: average ΔV_j for event type e_i. This indicates how valuable/impactful e_i is in winning a game.
    - $N_i = |j: E(j) = e_i|$: number of points ended with event type e_i
    - $\sum_{j:E(j)=e_i} \Delta V_j$: Total ΔV_j over all points ended with event type e_i

An Example

Point j	Event $E(j)$	ΔV_j
1	ace/winner (e_1)	+0.1
2	winner (e_3)	+0.08
3	double fault (e_2)	–0.11
4	ace/winner (e_1)	+0.12

- Assume $V(s)$ has been computed for each game state s
- Points where $E(j)=e_1$:
  - j=1,$\Delta V_1 = +0.1$
  - j=4,$\Delta V_4 = +0.12$
- So,
  - $N_1=2$
  - $w(e_1)=(0.10+0.12)/2=0.11$

Step 3: Player's Overall Performance

- Pick up a particular player.
- Prepare a point-by-point dataset from the player's matches.
- Compute: $\mathrm{X} = \frac{1}{N}\sum_{i=1}^{N} w(e_i)C_i$
  - C_i: count of the player's points classified as event type e_i
  - *N*: Total number of points recorded for the player (total number of classified points for the player)

- This metric
  - Measures the average impact of a player’s point outcomes on their probability of winning a game.
    - A higher X means the player consistently produces events that increase their chances of winning games.
    - A lower X means the player’s actions are less effective (or more harmful).
  - Goes beyond simple stats like “number of winners” and “number of errors”
    - Because it accounts for
      - Game context (e.g. 30-30 v.s. 40-0)
      - True impact on winning games
      - Both positive and negative events
  - Measures a player’s overall performance relative to the player population represented in the dataset used.
    - If weights (w) are computed based on the data from all WTA matches in 2025, for example, this metric measures overall performance relative to the WTA player population.

Appendix
(Ignore this section)

- Given a set of event types: $\mathcal{E} = \{e_1, e_2, \ldots e_i, \ldots e_N\}$
- For every point $j = 1, 2, \ldots, M$ in the dataset,
  - Identify the pre-point game state $s_{j,\text{before}}$
  - Identify the post-point game state $s_{j,\text{after}}$
  - Identify the event type for the server: $E(j, \text{server}) \in \mathcal{E}$
  - Identify the event type for the returner: $E(j, \text{returner}) \in \mathcal{E}$
  - Compute $\Delta V_j^{\text{server}} = V(s_{j,\text{after}}) - V(s_{j,\text{before}})$
    - $\Delta V_j^{\text{server}}$ indicates how much $E(j, \text{server})$ increased the server's chance of winning (i.e. how valuable/important $E(j, \text{server})$ is).
  - Compute $\Delta V_j^{\text{client}} = V(s_{j,\text{before}}) - V(s_{j,\text{after}})$
    - $\Delta V_j^{\text{client}}$ indicates how much $E(j, \text{client})$ increased the client's chance of winning (i.e. how valuable/important $E(j, \text{client})$ is).
    - Note that $\Delta V_j^{\text{client}} = -\Delta V_j^{\text{server}}$
  - Assign $\Delta V_j^{\text{server}}$ to the event type $E(j, \text{server})$
  - Assign $\Delta V_j^{\text{client}}$ to the event type $E(j, \text{client})$

An Example

Point j	**Perspective** p	**Event** $E(j, p)$	ΔV_j^p
1	server	ace (e_1)	+0.1
1	returner	opponent ace (e_3)	–0.1
2	server	winner (e_2)	+0.08
2	returner	opponent winner (e_4)	–0.08
3	server	ace (e_1)	+0.11
3	returner	opponent ace (e_3)	–0.11

- Assume $V(s)$ has been computed for each game state s
- For each point, make 2 rows (for server and returner)

- Then, compute the average ΔV_j^p for each event type
  - Formally, $w(e_i) = \frac{1}{N_i}\sum_{j:E(j,p)=e_i} \Delta V_j^p$
    - $w(e_i)$: average ΔV_j^p for event type e_i. This indicates how valuable/impactful e_i is in winning a game.
    - $N_i = |j: E(j,p) = e_i|$: number of points with event type e_i
    - $\sum_{j:E(j,p)=e_i} \Delta V_j^p$: Total ΔV_j^p over all points with event type e_i

An Example

Point j	Perspective p	Event $E(j, p)$	ΔV_j^p
1	server	ace (e_1)	+0.1
1	returner	opponent ace (e_3)	–0.1
2	server	winner (e_2)	+0.08
2	returner	opponent winner (e_4)	–0.08
3	server	ace (e_1)	+0.11
3	returner	opponent ace (e_3)	–0.11

- Points where $E(j, p)=e_1$:
  - j=1, $\Delta V_1^{\text{server}} = +0.1$
  - j=3, $\Delta V_3^{\text{server}} = +0.11$
- So,
  - $N_1=2$
  - $w(e_1)=(0.10+0.11)/2=0.105$

Step 3: Player's Overall Performance

- Pick up a particular player.
- Prepare a point-by-point dataset from the player's matches.
- Compute: $\mathrm{X} = \frac{1}{N}\sum_{i=1}^{N} w(e_i)C_i$
  - C_i: count of the player's points classified as event type e_i

- This metric
  - Measures the average impact of a player’s point outcomes on their probability of winning a game.
    - A higher X means the player consistently produces events that increase their chances of winning games.
    - A lower X means the player’s actions are less effective (or more harmful).
  - Goes beyond simple stats like “number of winners” and “number of errors”
    - Because it accounts for
      - Game context (e.g. 30-30 v.s. 40-0)
      - True impact on winning games
      - Both positive and negative events
  - Measures a player’s overall performance relative to the player population represented in the dataset used.
    - If weights (w) are computed based on the data from all WTP matches in 2025, for example, this metric measures overall performance relative to the WTP player population.